✍️ Exam Answer

Predictive maintenance reduces equipment breakdown losses by continuously monitoring machine conditions using sensors and data analysis. It detects abnormal conditions and predicts possible failures before they occur. This helps in timely maintenance, reducing unplanned downtime, repair cost and production losses.

⭐ Keywords

Remember:

Monitor → Predict → Maintain → Reduce Breakdown

✍️ Exam Answer — 2 Marks

Two benefits of minimizing the Six Big Losses using Industry 4.0 are:

1. **Increased productivity:** IoT and sensors reduce machine downtime and improve equipment utilization.
2. **Improved quality:** Sensors and data analysis help detect defects early and reduce defective products.

🧠 Remember:

Losses ↓ → Productivity ↑ + Quality ↑

✍️ Exam Answer

Conventional Machine	Smart Machine
Operates mainly using fixed/manual control.	Uses sensors and intelligent control.
Has limited monitoring and feedback.	Provides continuous real-time monitoring and feedback.

⭐ Memory Trick

Conventional = Fixed + Manual

Smart = Sense + Connect + Decide

Remember this! 🔥

✍️ Exam Answer — 2 Marks

A smart machine can improve a CNC manufacturing process by using sensors to monitor temperature, vibration and tool condition. The data is analyzed in real time to detect abnormalities and take corrective action. This reduces downtime, improves quality and increases productivity.

🧠 Remember:

Sensors → Monitor → Detect → Improve

✍️ Exam Answer — 2 Marks

Two important features of a smart machine are:

1. **Real-time sensing:** Sensors continuously measure machine parameters such as temperature and vibration. It helps in quick fault detection.
2. **Connectivity:** The machine can communicate data through a network. It enables data sharing and remote monitoring.

🧠 Remember:

Smart Machine = S + C

S = Sensing

C = Connectivity

==========set 2==========

✍️ Exam Answer

Industry 4.0	Industry 5.0
Focuses on automation, IoT, AI and smart machines.	Focuses on collaboration between humans and intelligent machines.
Main goal is higher efficiency and productivity.	Main goal is human-centric, sustainable and resilient manufacturing.

🧠 Memory Trick

4.0 = Automation

5.0 = Human + Machine

✍️ Exam Answer

IoT uses sensors and connected devices to continuously monitor machines and manufacturing processes. It detects abnormal conditions early, helping to prevent breakdowns, reduce downtime, minimize defects and improve production efficiency.

🧠 Remember:

IoT → Monitor → Detect → Reduce Loss